

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S8 (R, S) Exam April 2025 (2019 Scheme)

Course Code: CXT402**Course Name: ADVANCED COMPUTER GRAPHICS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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|----|---|-----|
| 1 | Define a Sweep. | (3) |
| 2 | Write the properties of a Bezier curve. | (3) |
| 3 | Draw the 3D viewing pipeline. | (3) |
| 4 | What is isometric projection? | (3) |
| 5 | What is an illumination model? | (3) |
| 6 | What is intensity attenuation? | (3) |
| 7 | What are the three main characteristics of light? | (3) |
| 8 | What is XYZ color model? | (3) |
| 9 | Explain about conventional animations. | (3) |
| 10 | Explain raster animation. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) Explain how polygon surfaces are specified? | (9) |
| | b) Write any two 3D display methods. | (5) |

OR

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|----|--|-----|
| 12 | a) Describe the term Bezier curves. | (7) |
| | b) What is a spline? Explain with continuity conditions on a spline. | (7) |

Module II

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| 13 | a) Explain basic 3D transformations. | (14) |
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OR

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| 14 | a) Explain reflections, shears and composite transformations | (9) |
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- b) Given a 3D triangle with points (1,2,3), (4,5,6) and (7,8,9) undergoes a shear transformation in Y axis with shearing parameter 2 on X axis, 1 on Y axis and 3 on Z axis and find out the new coordinates of the object. (5)

Module III

- 15 a) Explain basic illumination models. (14)

OR

- 16 a) What is surface rendering? Explain any two surface rendering methods. (7)
b) Explain basic Ray-Tracing Algorithm with diagram. (7)

Module IV

- 17 a) Explain various color models with necessary diagrams and equations. (14)

OR

- 18 a) Explain HSV and HLS color model. (7)
b) Write the conversion between HSV and RGB models. (7)

Module V

- 19 a) How Morphing methods can be applied to any motion or transition involving a change in shape? Explain with examples. (14)

OR

- 20 a) Explain different ways to specify motion in animation systems. (9)
b) What is Interpolation? Explain its types. (5)
